

Scholastic Aptitude Test, 2016 EC, natural science, entrance exam, round 1, Tigray

Grade 12 · Aptitude · 2016 E.C.

Time allowed: 120 minutes

Questions: 39

Total marks: 39

Name _____ Grade _____ ID No. _____ Date _____

1. INQUISITIVE (1)

A. curious	B. uninterested
C. assertive	D. intrusive
2. INDUSTRIOUS (1)

A. innovative	B. antisocial
C. productive	D. lazy
3. METICULOUS (1)

A. clueless	B. careless
C. dangerous	D. miraculous
4. DEVASTATE (1)

A. injure	B. frighten
C. shock	D. renovate
5. PRICELESS (1)

A. valueless	B. useful
C. worthless	D. invaluable
6. LAUGHTER: DELIGHT; _____: discontent (1)

A. satisfaction	B. sorrow
C. excitement	D. complaint
7. DESCRIBE: PLACE; _____: idea (1)

A. tell	B. think
C. give	D. explain
8. APPLE: FRUIT; _____: cereal (1)

A. crop	B. wheat
C. orange	D. juice
9. Common: _____; SLIGHT: SEVERE (1)

A. extraordinary	B. usual
C. expected	D. dull
10. KITCHEN: HOUSE; _____: school (1)

A. teacher	B. desk
C. classroom	D. director

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11. People flock to the capital city in large numbers in search of opportunities. This has created problems in the city as limited resources have to be shared with the new arrivals. It's becoming harder to find jobs in the city, and house rent has skyrocketed. (1)
- A. The city administration should regulate entrance into the capital city.
B. More opportunities should be created in the migrants' original areas.
C. The city administration should create more jobs and construct more condominiums.
D. The city administration should look for international aid to create jobs for the new arrivals.
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12. Most people think that education is crucial for success in an individual's life. There are a number of people who have applied their education to improve their lives. However, there are also those who have very little education who have become multi-billionaires. We can, therefore, safely conclude that: (1)
- A. formal education is a waste of time.
B. uneducated people who have become successful are always corrupt.
C. formal education helps people to improve their talents.
D. Becoming successful in life is just a matter of chance.
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13. Five athletes: G/Egziabher, Haile, Kenenisa, Solomon, and Sileshi took part in a ten-thousand-meter competition. Sileshi finished the competition before Kenenisa and Solomon but after G/Egziabher. Haile finished before Kenenisa but after Solomon. Who finished third? (1)
- A. Kenenisa
B. Sileshi
C. G/Egziabher
D. Solomon
-
14. Barack Obama was a popular American President. He became president at the age of 46. He is black American. Biden has not been as popular as Obama in his presidency. He is a white, 88-year-old American. Nelson Mandela was also a popular South African president at the age of 88. Which one of the following statements is acceptable? (1)
- A. Black people can make better presidents.
B. American people are free from racism.
C. Age and leadership quality are not necessarily interrelated.
D. Leading America is more demanding than leading South Africa.
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15. Of the three models of cars (A, B, and C), Model A costs slightly lower than Model B. Model C is slightly less expensive than Model A but its fuel consumption is slightly higher than both Model A and Model B. Model B consumes less fuel per kilometer than Model A but it is not as efficient as Model C. (1)
- A. Model B is the least expensive.
B. Model C is the most expensive.
C. Model A consumes the least fuel.
D. Model B is the most expensive.
-
16. Sara hates people who tell lies the most. (1)
- A. Sara hates people who tell lies most of the time.
B. What Sara hates more than anything is liars.
C. Sara has no problem with people except liars.
D. Sara always tells the truth.
-
17. Most French people are friendly and fair. Nicola is French. I also know some French people who are biased. Therefore, (1)
- A. may be friendly
B. is both friendly and fair
C. cannot be biased
D. must be fair

18. The concept of democracy is so _____ that it conveys variable meanings to different people. (1)

A. concrete
B. abstract
C. divisive
D. advanced

19. As more people settle on areas previously inhabited by wild animals, most of such animals become _____. (1)

A. frightened
B. evacuated
C. domesticated
D. endangered

20. Undue competition among people for natural resources will lead to _____ of land. (1)

A. innovation
B. overpopulation
C. degradation
D. democratization

21. If you want your country to register economic development, it is _____ that you deal with corruption first. (1)

A. imperative
B. arguable
C. possible
D. regrettable

22. Everyone respects the manager for his _____. (1)

A. integrity
B. strength
C. charisma
D. elegance

23. How many sheep are there if all the hens and sheep in the 500 legs and 200 heads? (1)

A. 50
B. 100
C. 150
D. 80

24. Which one of the following points is the intersection of the graphs of the lines $x - y + 1 = 0$ and $-x - y + 2 = 0$? (1)

A. $(-\frac{1}{2}, -\frac{3}{2})$
B. $(\frac{1}{2}, \frac{3}{2})$
C. $(\frac{3}{2}, \frac{1}{2})$
D. $(-\frac{3}{2}, -\frac{1}{2})$

25. A school football team includes only grade 10, grade 11, and grade 12 students, in a ratio 1:2:3. How many grade 12 students are in the team if the team has 30 members? (1)

A. 20
B. 15
C. 10
D. 5

26. Which of the following ordered pairs is included in the solution set of the system of linear inequalities given by $-2x + y \geq 0$ and $3x + 2y \leq 6$? (1)

A. (3, 1)
B. (1, -2)
C. (1, 4)
D. (-1, 1)

27. What is the coefficient of x^3 in the expansion of the product of $(x^3 - 20x^2 + 4x + 2)(15x^3 + x^2 - 2x - 1)$? (1)

A. 73
B. 93
C. 83
D. 103

28. When the polynomial function $f(x) = x^4 - 6x^3 + 15x^2 - 18x + d$ is divided by $g(x) = x^2 - 2x + 3$, the quotient is $x^2 - bx + 4$ and the remainder is $cx - 2$, where b, c and d are constants. Then what is the value of c ? (1)

A. 10
B. -4
C. 4
D. 2

29. A television was on sale for 20% decrease from its original price. What was the original price of the television if the sale price was Birr 40,000 tax-free? (1)
- A. Birr 54,000
B. Birr 44,000
C. Birr 50,000
D. Birr 33,000
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30. A man throws a rock in a vertical upward trajectory with velocity of 7 m/s on 90 meters tall roof of his house. Its height can be modeled using the quadratic function $h(t) = -5t^2 + 7t + 90$. At what approximate time does the rock hit the ground? (1)
- A. 3 second
B. 4.7 second
C. 6.7 second
D. 5 second
-
31. Let $f\left(\frac{2}{3}x - \frac{4}{3}\right) = 5x^3 + 16x^2 + 2x + 1$, and x be a real number. Consider the two quantities A and B given below. Quantity A Quantity B f(2) 600 (1)
- A. Quantity A is greater than Quantity B.
B. Quantity A is less than Quantity B.
C. The two quantities are equal.
D. A relationship cannot be determined.
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32. The following table shows the profit of two shops in the month of January and February in 2015 Ethiopian calendar. Shop January February Shop 1 Birr 20,000 Birr 30,000 Shop 2 Birr 10,000 Birr 15,000 Quantity A: Percentage increase in shop 1 Quantity B: Percentage increase in shop 2 (1)
- A. Quantity A is greater than Quantity B.
B. Quantity A is less than Quantity B.
C. The two quantities are equal.
D. The two quantities cannot be compared.
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33. What is the probability that the sum is neither 5 nor 10 if two dice are thrown? (1)
- A. $\frac{2}{9}$
B. $\frac{7}{36}$
C. $\frac{7}{9}$
D. $\frac{29}{36}$
-
34. On a total of 9 chairs, 4 boys and 5 girls have to be seated. If the boys and girls sit alternately, how many different ways may they be seated? (1)
- A. 5
B. 2,880
C. 126
D. 362,880
-
35. Which one of the following can be the first four terms of a geometric sequence whose fourth term is 125 and whose tenth term is $\frac{125}{64}$? (1)
- A. 1000, 500, 250, 125, ...
B. 1000, 500, -250, 125, ...
C. 3375, 1125, 375, 125, ...
D. -500, 375, -250, 125, ...
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36. Which term of the arithmetic sequence {24, 30, 36, 42, 48, 54, 60, ...} is 204? (1)
- A. 30th
B. 31st
C. 32nd
D. 33rd
-
37. Which one of the following is the solution set of the system of linear equations given by $3x - 4y = 2$ and $-6x + 8y = -10$? (1)
- A. {(2, 1)}
B. {(3, 1)}
C. {}
D. {(0, -2)}
-
38. Which of the following is the sum of the first 16 terms of the arithmetic sequence $\left\{-2, \frac{3}{2}, 5, \frac{17}{2}, \dots\right\}$? (1)
- A. 420
B. 380
C. 402
D. 388

39. What is the value of α if $\sum_{n=1}^{\infty} 3^{-\alpha n} = \frac{1}{2}$?

(1)

A. 1

B. -1

C. 2

D. -2

— END OF EXAMINATION —

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